MINI PROJECT PRESENTATION

AUTOCONNECT

ISP Automation System

Pranjal Rajesh Upadhyay

PRN: 1032233275 | Roll No: 39 | Panel: B

B.Tech CSE AIDS • Mini Project using JAVA Programming

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PROJECT OVERVIEW

AutoConnect is a console-based ISP (Internet Service Provider) Automation System built in Java using core Object-Oriented Programming principles. It simulates real-world ISP operations including customer management, service plans, billing, and complaint handling — all without any database or GUI.

Customer Management

Register & manage customers with auto IDs

Plan Assignment

3-tier plans: Basic, Standard, Premium

Billing System

Auto bill generation & payment tracking

Complaint Handling

Raise & track service complaints

Technology: Java SE • OOP • Collections • Stream API • Enums • Scanner

PROBLEM STATEMENT

❌ Manual System Problems	✅ AutoConnect Solution
• Paper-based customer records — no centralization	• Centralized in-memory customer management
• Manual billing leads to frequent errors	• Auto bill generation from plan price
• No automatic connection suspension logic	• Enum-based ACTIVE/SUSPENDED status control
• Complaints tracked via phone calls only	• Complaint list with OPEN/RESOLVED tracking
• No unique ID system for customers	• Auto-incremented unique customer IDs
• Slow data retrieval & no status tracking	• Stream API for instant data retrieval

SYSTEM ARCHITECTURE

UI LAYER

AutoConnect.java (main)

Console menu, Scanner input, switch-case routing

SERVICE LAYER

ISPService.java

Business logic: register, assign, bill, pay, complain

MODEL LAYER

Customer, Plan, Bill, Complaint, Enums

Data classes with encapsulated fields & getters

Clean separation of concerns — Model | Service | UI

OOP CONCEPTS USED

🔒 Encapsulation

All fields are private
Accessed via getters/setters
Data integrity maintained

🎭 Abstraction

ISPService hides complexity
Simple method calls for UI
Business logic separated

🔄 toString() Override

@Override toString()
Custom output format
Used in all model classes

📦 Collections

ArrayList for storage
Stream API for filtering
forEach for iteration

🎯 Enums

ConnectionStatus enum
BillStatus enum
Type-safe state values

🔁 Clean Loop

while(true) menu loop
Switch expressions (Java 14)
return to exit cleanly

KEY CODE SNIPPETS

Enum – Type-Safe Status

```
enum ConnectionStatus { ACTIVE, SUSPENDED
}
enum BillStatus { PAID, UNPAID }
```

Stream API – Customer Lookup

```
return customers.stream()
  .filter(c -> c.getId() == id)
  .findFirst().orElse(null);
```

Auto ID Generation

```
public void registerCustomer
  (String name, String phone) {
  customers.add(new Customer(
    customerIdCounter++, name, phone));}
```

Bill Generation Logic

```
Bill bill = new Bill(
  billIdCounter++, customerId,
  c.getPlan().getPrice());
bills.add(bill);
```

Auto-Activate on Plan Assign

```
public void setPlan(Plan plan) {
  this.plan = plan;
  this.status =
    ConnectionStatus.ACTIVE;
}
```

Modern Switch Expression

```
switch (choice) {
  case 1 -> {
    service.registerCustomer(...);
  }
  case 9 -> { return; }
}
```

SYSTEM FLOW

1	→	2	→	3	→	4	→	5
Register Customer		**Assign Plan**		**Generate Bill**		**Pay Bill**		**Raise Complaint**
Name + Phone → Auto ID		Select Plan → ACTIVE		Plan Price → UNPAID		Bill ID → PAID		Issue text → OPEN

⚡ **AUTOMATION RULES:** No Plan → SUSPENDED Status | Plan Assigned → ACTIVE | Bill Paid → PAID Status | Complaint Raised → OPEN

SAMPLE OUTPUT

▶ Main Menu

```
===== AutoConnect ISP System =====
1. Register Customer
2. View Customers
3. Show Plans
4. Assign Plan
5. Generate Bill
6. Pay Bill
7. Raise Complaint
8. View Complaints
9. Exit
Enter choice: _
```

▶ Sample Session

```
Enter choice: 1
Name: Pranjal
Phone: 9876543210
Customer registered!

Enter choice: 3
1. Basic | 50 Mbps | Rs.499
2. Standard | 100 Mbps | Rs.799
3. Premium | 300 Mbps | Rs.1299

Enter choice: 5
Customer ID: 1
Bill: 1 | Rs.799 | UNPAID

Enter choice: 6
Bill ID: 1
Payment successful!
```

Console-based • No GUI • No Database • Pure Java OOP

ADVANTAGES & FUTURE SCOPE

✅ Advantages

- ✓ Zero external dependencies
- ✓ Clean 3-tier OOP architecture
- ✓ Auto ID generation system
- ✓ Enum-based type-safe states
- ✓ Stream API for clean filtering
- ✓ Runs on any OS with JDK

🚀 Future Scope

- → MySQL / JDBC Database integration
- → JavaFX / Spring Boot UI layer
- → Online Razorpay payment gateway
- → Login & role-based auth system
- → Auto bill suspension scheduler
- → Reporting & analytics dashboard

CONCLUSION

AutoConnect demonstrates the power of Java OOP

From customer registration to billing and complaints — every operation follows clean OOP principles with separated Model, Service, and UI layers. This project bridges theoretical knowledge with real-world application design.

- ✓ Clean 3-tier architecture
- ✓ Enum-based state management
- ✓ Stream API integration
- ✓ Auto ID generation
- ✓ No external dependencies

THANK YOU

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